

Configuration-driven experimental pipeline

one YAML defines the run; the same runner produces evaluation artifacts for the paper

one experiment = one YAML config

Backbone

Qwen2-VL 2B/7B, BLIP-2

Dataset

DocVQA, ChartQA, ScienceQA, A-OKVQA, VQAv2

Objective

zero-shot, answer-only, rationale CE, expl.-aware, contrastive

Metrics

accuracy, ANLS, ROUGE/BLEU, drift

Experiment runner

builds model, dataset, objective, collator, and metrics
verifies target masking before training

Baseline eval

score the frozen model

Task fine-tune

Qwen2-VL QLoRA or
BLIP-2 Q-Former path

Final eval

re-score the adapter

Answer

exact / multiple-choice
ANLS / relaxed accuracy

Explanation

ROUGE-L, BLEU
only for rationale runs

Evidence sensitivity

top / random / blank masks
mask → answer drift

Saved artifacts

resolved config, adapter, generations, results JSON/CSV

Paper outputs

tables, curves, transfer matrices, qualitative panels